NAGIZAAZ SHAIK

GENERATIVE AI ENGINEER · AGENTIC AI · LLMS · RAG · AI SAFETY · AWS ML

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Generative AI and agentic AI engineer who ships production LLM, RAG and agent systems and studies how they fail. Cut ML deployment failures 85% and compliance research from about 2 hours to under 20 minutes at two regulated credit unions. Doctoral researcher in AI-agent security; 20 merged PRs across 18 open-source projects.

SKILLS

Generative AI: Claude, OpenAI, Hugging Face, Qwen2.5, LoRA / PEFT, prompt engineering, structured outputs, vLLM, Ollama

Agentic AI: LangGraph, CrewAI, MCP, tool calling, multi-agent orchestration, agent memory, guardrails

Deep Learning & ML: PyTorch, Hugging Face Transformers, BERT, LoRA fine-tuning, YOLO11 (computer vision), PPO (reinforcement learning), XGBoost, scikit-learn

RAG Pipelines: LangChain, embeddings, Pinecone, FAISS, OpenSearch, BM25 hybrid search, reranking, HyDE, corrective RAG

AWS: Bedrock, SageMaker (endpoints, Pipelines, Model Monitor), Lambda, S3, Glue, Redshift, EKS, IAM, CloudWatch

Eval & Safety: RAGAS, LLM-as-judge, held-out tests, OWASP LLM Top 10 red-teaming, prompt-injection tests, PII redaction

MLOps & Data: Databricks, MLflow, Airflow, Kafka, FastAPI, Docker, Kubernetes, Terraform, ArgoCD, GitHub Actions, n8n

Languages & Domain: Python, SQL, Go | credit-risk modeling, regulatory compliance, SHAP

EXPERIENCE

ML Platform Engineer · Contract | Regulated credit union *Jul 2025 – Jun 2026*

- Built the credit union's internal GenAI gateway (Claude, OpenAI, Ollama) with evaluation guardrails, cost controls, and RAGAS grounding checks: **one governed path** to ship GenAI features.
- Founded the AWS ML platform: SageMaker endpoints and Pipelines, MLflow registry with gated promotion and automated rollback, per-environment VPC isolation; **deployment failures down 85%** in 60 days.
- Owned ML data security (AES-256 at rest and in transit, least-privilege IAM, audit logging); passed **two consecutive regulatory audits with zero findings**.
- Codified the infrastructure in Terraform with GitOps on ArgoCD, making every environment change reproducible and peer-reviewed.
- Mentored two engineers into ML and platform ownership through weekly architecture and code reviews.

Machine Learning Engineer · Contract | Regulated credit union *Jul 2024 – Jun 2025*

- Built a RAG pipeline over 500+ internal policy and regulatory documents (LangChain, vector search, Claude); compliance research went **from ~2 hours to under 20 minutes** per question.
- Fine-tuned a 14-class BERT intent classifier to fix member-support misrouting, served on FastAPI at **sub-200 ms p99**.
- Built the first-pass consumer-loan underwriting model (XGBoost on SageMaker) as the first dedicated ML engineer, keeping approval rates within risk thresholds.
- Replaced a manual multi-hour retraining cycle with a weekly Airflow pipeline: rolling-window retraining, holdout evaluation, and AUC / precision / calibration promotion gates.
- Set up SageMaker Model Monitor drift detection with custom baselines; it caught **two data-drift events** before they reached credit decisions.

Senior Data Engineer | IT services firm (financial-services client) *Jul 2020 – Jun 2023*

- Led the Oracle 11g to Amazon Redshift migration for five business units, redesigning the schema for columnar storage and rebuilding 20+ stored procedures as Airflow DAGs; **query runtime down 60%**.
- Built 15 production ETL/ELT pipelines with a multi-stage validation layer that caught a silent upstream corruption affecting three months of financial reports.
- Promoted from Associate to Senior in two years; led code reviews for three engineers and was primary client contact on two accounts.

Python Intern | Technology company *Jan 2019 – Oct 2019*

- Automated a 4-hour weekly manual ETL job as a scheduled Python job with anomaly checks on 7-day baselines; earned a return-internship offer.

RESEARCH

Doctoral Researcher, Applied AI | Belhaven University *Aug 2026 – Present*

- Found a sandbox bypass in the code executor of a widely used AI-agent framework and reported it through a GitHub security advisory; **a downstream project adopted my proposed fix** and credited the report (working paper *When Vendoring Defeats the Patch*).
- Released guard-gap-detector, a read-only static checker with a synthetic corpus, so the finding can be reproduced without third-party code or an exploit.
- Showed that a planted label pushes a LoRA-tuned financial classifier to the attacker's label on **up to 92%** of items (five markets, three seeds, controls); a standard confidence gate missed about half of one attack's flips (working paper *Fragile in Different Ways*).

PROJECTS

Finance Agent Crew | Python, asyncio, Claude, FastAPI github.com/shaikn6/finance-agent-crew

A ticker goes in, a sourced equity-research brief comes out: concurrent SEC EDGAR, market, and news collection feeds parallel fundamental, sentiment, and risk agents, and a writer agent returns a typed BUY / HOLD / SELL report; a failed data source is recorded, not fatal.

LLM Gateway | Python, FastAPI, Redis, Docker github.com/shaikn6/llm-gateway

One OpenAI-compatible API over Anthropic, OpenAI, and Ollama with per-tenant caching, rate limiting, A/B routing, cost tracking, and audit logs; contract tests catch breaking provider SDK changes in CI.

Financial Text Classification with LoRA | PyTorch, PEFT, Qwen2.5, FinBERT github.com/shaikn6/fin-lora

A LoRA-tuned Qwen2.5-0.5B trained on tweets scored only 59.0% on news; retrained on news, tweets, and headlines, it reached **84.7%** vs 71.7% for TF-IDF and 73.6% for FinBERT (267-example news set, about ±5 points). A confidence cascade that sends only uncertain items to the LLM cut average latency **63%** for 0.4 points of accuracy.

lethe: Agent Memory Server | Go, Python, LangChain, MCP github.com/shaikn6/lethe

Learns from feedback which memories to return to an agent. The first learner lost to a hand-set rule (0.611 vs 0.641 evidence in the top 10) because new memories were never tried; forced exploration fixed it: **0.707** on held-out LoCoMo conversations, at 7,119 requests/s.

sigdet: Signature Detection with YOLO11 | PyTorch, Ultralytics YOLO11 github.com/shaikn6/sigdet

Leakage-aware split (110/26/26 images, hash-checked); over 3 seeds the small model scores **0.939 ± 0.005** test mAP50-95 and the larger one 0.859 ± 0.041, so smaller was better and more stable. Includes a corruption robustness study.

exec-rl: Trade Execution with PPO | PyTorch, PPO github.com/shaikn6/exec-rl

From-scratch PPO agent vs TWAP and Almgren-Chriss on 20,000 held-out simulated paths, 3 seeds: **0.22 bps** shortfall vs 18.8 bps for both baselines with the signal, no better than TWAP without it. Simulation only.

LLM Safety Auditor | Python, sentence-transformers, FastAPI github.com/shaikn6/llm-safety-auditor

Red-teaming harness that scores **50 curated adversarial attacks** (350 with mutations) against the OWASP LLM Top 10, with a 6-layer detector and PDF audit reports; 635 tests.

Open Source | 20 merged PRs across 18 projects, incl. crewAI (60K stars, incl. a fix for a bug I reported), MLflow (28K), Ray (44K), DSPy github.com/shaikn6

EDUCATION & CERTIFICATIONS

Belhaven University — Doctor of Business Administration (DBA), Applied AI · GPA 4.0 / 4.0 Aug 2026 – Present

University of Dayton — M.S. Computer Science · GPA 3.93 / 4.0 Aug 2023 – May 2025

CMR Engineering College — B.Tech. Computer Science and Engineering · GPA 8.9 / 10 Aug 2016 – May 2020

AWS Certified Machine Learning – Specialty • AWS Certified Data Engineer – Associate • GCP Professional Data Engineer